CS254 MINI PROJECT REPORT

OASYS

A Multi-Sector Enterprise Operations and Analytics System

Submitted by:

DHARAVATH SAMPATH (241CS122)
POLURI SAI JISHNU (241CS140)
SORNALA SUHITHA (241CS156)
UTKOOR VENKATESH (241CS161)

Department of Computer Science and Engineering

National Institute of Technology Karnataka, Surathkal

5th April 2026

CONTENT PAGE

1. Introduction 3
2. Literature Survey 4
3. Project Objectives 5
4. Features Included 5
5. Limitations 6
6. Requirement Specification
   6.1 Frontend 6
   6.2 Backend 7
7. System Design
   7.1 Tables Mentioned 7
   7.2 ER Diagram 8
   7.3 Database Schema 9
8. Implementation
   8.1 Tools and Technologies Used 10
   8.2 Authentication Flow 10
9. References 11

INTRODUCTION

The project "OASYS" (Operational Analytics System) is a full-stack Database Management System (DBMS) application designed to simulate the operations and analytics of large-scale enterprises. In modern corporate environments, organizations operate across multiple domains such as finance, telecommunications, logistics, energy, and manufacturing. Managing such diversified operations requires a centralized system that ensures data integration, real-time monitoring, and efficient decision-making. OASYS addresses this need by providing a unified platform that consolidates multiple business verticals into a single database-driven system.

The system consists of six major domains: Fintech, Real Estate, Telecom, Logistics, Energy, and Automobiles. Each domain maintains its own operational data while being integrated into a centralized database architecture. This enables stakeholders to gain both a high-level overview and domain-specific insights through dashboards and analytical reports. The platform focuses on structured data management, scalability, and performance optimization.

OASYS demonstrates the practical implementation of key DBMS concepts such as normalization, indexing, views, stored procedures, triggers, and transactions. These concepts ensure data integrity, consistency, and efficient query processing. The system is built using a three-tier architecture, comprising a frontend for user interaction, a backend for business logic, and a database layer for data storage and processing.

Additionally, the application supports real-time analytics, operational workflows, and report generation with downloadable outputs. By integrating analytics with transaction management, OASYS enhances business intelligence and supports data-driven decision-making.

Overall, the project highlights how modern database systems can be used to design scalable, reliable, and efficient enterprise-level solutions.

LITERATURE SURVEY

The development of OASYS is based on the study of existing enterprise systems, domain-specific platforms, and traditional data management approaches. The following systems and practices form the foundation for understanding the need for a unified solution:

1. **Enterprise Resource Planning (ERP) Systems**

   ERP systems provide centralized management of multiple business processes such as finance, supply chain, and operations within a single platform. They ensure data consistency and integration but are often complex, expensive, and not easily customizable for smaller or academic implementations.

2. **Domain-Specific Management Systems**

   Various systems exist for specific domains such as banking applications, telecom management systems, logistics tracking platforms, and energy billing systems. While these systems efficiently handle operations within their domain, they lack cross-domain integration, resulting in fragmented data and limited overall visibility.

3. **Online Platforms and Dashboards**

   Modern platforms provide dashboards and analytical tools for monitoring performance metrics. However, these are typically limited to a single domain and do not support unified analytics across multiple business units.

4. **Traditional Data Management Methods**

   Many organizations still rely on spreadsheets, manual records, or standalone applications for managing operational data. These methods lead to redundancy, inconsistency, and inefficient data retrieval, making them unsuitable for large-scale enterprise operations.

5. **Lack of Integrated Analytics Systems**

   Existing solutions often fail to combine operational data with analytics in a single system. This limits the ability of organizations to perform real-time monitoring and data-driven decision-making across multiple domains.

The analysis of these existing systems highlights the need for a centralized, scalable, and integrated platform. OASYS is designed to address these limitations by providing unified data management, cross-domain analytics, and efficient operational control within a single system.

PROJECT OBJECTIVES

The primary objective of OASYS (Operational Analytics System) is to create a centralized platform that integrates multiple business domains into a single system for monitoring, analysis, and operational control. It simplifies data management while enabling efficient decision-making through real-time analytics and reporting.

- Develop a centralized database for managing multi-domain business data.
- Enable unified access to different business verticals through a single dashboard.
- Provide real-time analytics and visualization of operational data.
- Simplify transaction processing and operational workflows.
- Ensure data integrity and consistency using relational database design.

FEATURES INCLUDED

OASYS integrates multiple features to improve usability and efficiency for managing enterprise operations across domains.

- Centralized dashboard with cross-vertical analytics.
- Domain-specific modules for Fintech, Real Estate, Telecom, Logistics, Energy, and Automobiles.
- Real-time data updates and visualization using charts and tables.
- Transaction-based operations such as fund transfers and order processing.
- Automated audit logging for tracking system activities.
- Report generation with downloadable PDF support.
- Role-based access through a single admin control system.

LIMITATIONS

The current systems operate within certain boundaries:

- Existing systems operate within individual domains without unified integration.
- Real-time synchronization across different modules is limited.
- Advanced analytical insights across domains are minimal.
- Scalability across multiple business verticals is constrained.
- Handling large-scale multi-domain data efficiently is challenging.
- Decision-making support through integrated analytics is restricted.
- Flexibility in extending the system to new domains is limited.
- Performance optimization across multiple datasets can be improved.
- End-to-end workflow integration across domains is not fully streamlined.

Addressing these aspects can significantly enhance the system's scalability, performance, and integration capabilities, enabling it to be effectively utilized in large-scale enterprise environments.

REQUIREMENT SPECIFICATION

6.1 Frontend Requirements

- Admin Login Page (Username and Password fields)
- Centralized Dashboard to display cross-domain analytics and KPIs
- Navigation interface to access different business verticals
- Data visualization using charts (bar, line, donut) and tables
- Forms for operations such as fund transfer, order placement, and subscription renewal
- Filters for selecting date ranges, verticals, and report parameters
- Report generation interface with option to download as PDF
- Real-time data updates through periodic refresh

6.2 Backend Requirements

- Database: MySQL
- Server: Node.js with Express.js framework
- RESTful APIs for analytics, transactions, reporting, and data retrieval
- Execution of stored procedures for operations like fund transfer, order placement, and subscription renewal
- Trigger-based audit logging for tracking system activities
- Transaction management to ensure data consistency and rollback on failure
- Query optimization using indexes for efficient data retrieval
- Centralized handling of multi-domain data operations

SYSTEM DESIGN

7.1 Tables Mentioned

TABLE	PURPOSE
Verticals	Stores all business domains (Fintech, Telecom, etc.)
Users	Stores user details linked to specific domains
Audit_Logs	Records all system activities and operations
Bank_Accounts	Stores account details and balances for users
Transactions	Records all financial transactions (credit/debit/transfer)
Loans	Stores loan details including amount, interest, and status
Properties	Stores real estate listings with type, price, and status
RE_Deals	Records property deals with buyer, agent, and value
Telecom_Plans	Stores plan details such as type, data, and pricing
Subscriptions	Tracks user subscriptions with start and end dates
Network_Usage	Records daily usage data such as data and call minutes
Shipments	Stores shipment details including origin, destination, and status
Logistics_Routes	Stores route information such as distance and duration
Energy_Connections	Stores user energy connection details
Energy_Bills	Records billing details such as units consumed and payment status
Vehicles	Stores vehicle inventory with type, price, and stock
Vehicle_Orders	Records vehicle purchases and order status
Service_Requests	Stores service requests for vehicles and their status

7.2 ER Diagram

The ER diagram for OASYS represents the relationships between all entities across multiple business domains. The diagram is created using MySQL Workbench and illustrates the core tables, domain-specific tables, and their relationships, including primary and foreign key constraints.

- Users and Verticals form the core structure, linking all domain-specific tables across the system.
- One-to-many relationships exist between Users and entities such as Bank_Accounts, Subscriptions, Shipments, and Vehicle_Orders.
- Domain-specific relations include Properties → RE_Deals, Subscriptions → Network_Usage, and Energy_Connections → Energy_Bills.
- Audit_Logs is connected to all major tables, capturing system-wide activities for tracking and monitoring.

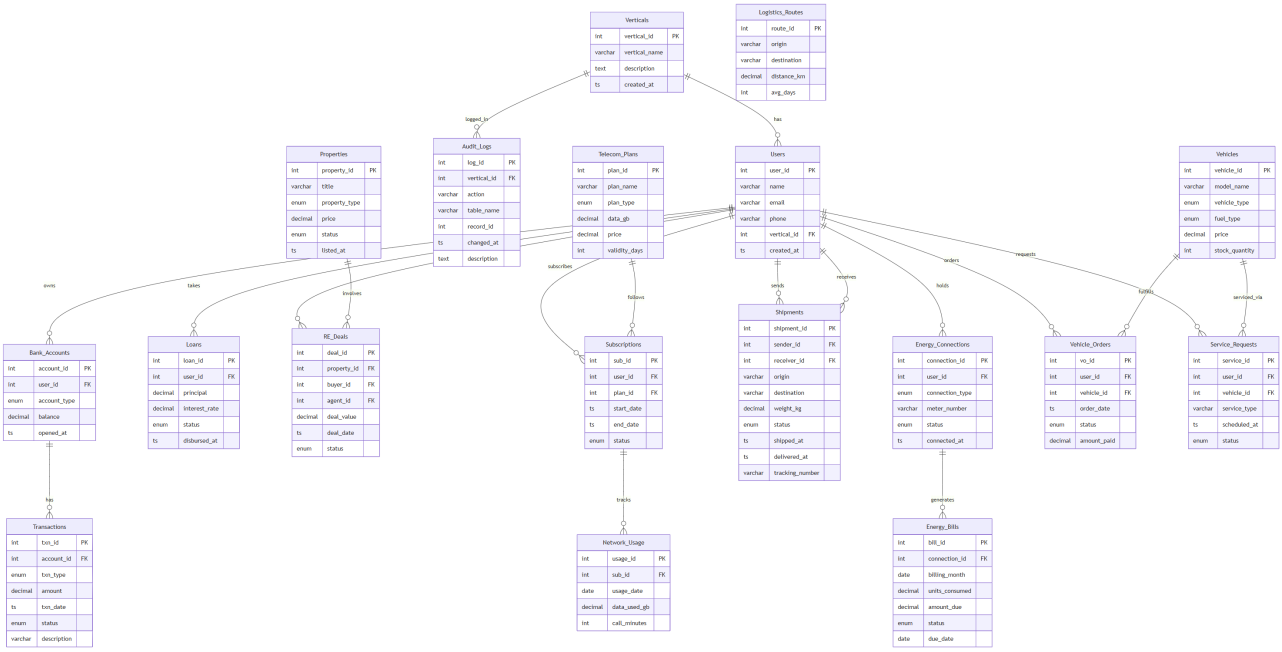

Figure 1: Entity-Relationship Diagram—OASYS

7.3 Database Schema

The database schema for OASYS defines how data is organized and managed across multiple business domains within a single system. It is designed using MySQL Workbench and includes shared core tables along with domain-specific tables, connected through well-defined primary and foreign key relationships.

- The Users table acts as the central entity, connecting to multiple domain tables such as accounts, subscriptions, shipments, and orders.
- Each business vertical maintains its own set of related tables, ensuring modular design while still being part of a unified schema.
- Relationships between tables support structured data flow, such as accounts to transactions and connections to billing records.
- The Audit_Logs table records key operations across all modules, enabling consistent tracking and system monitoring.

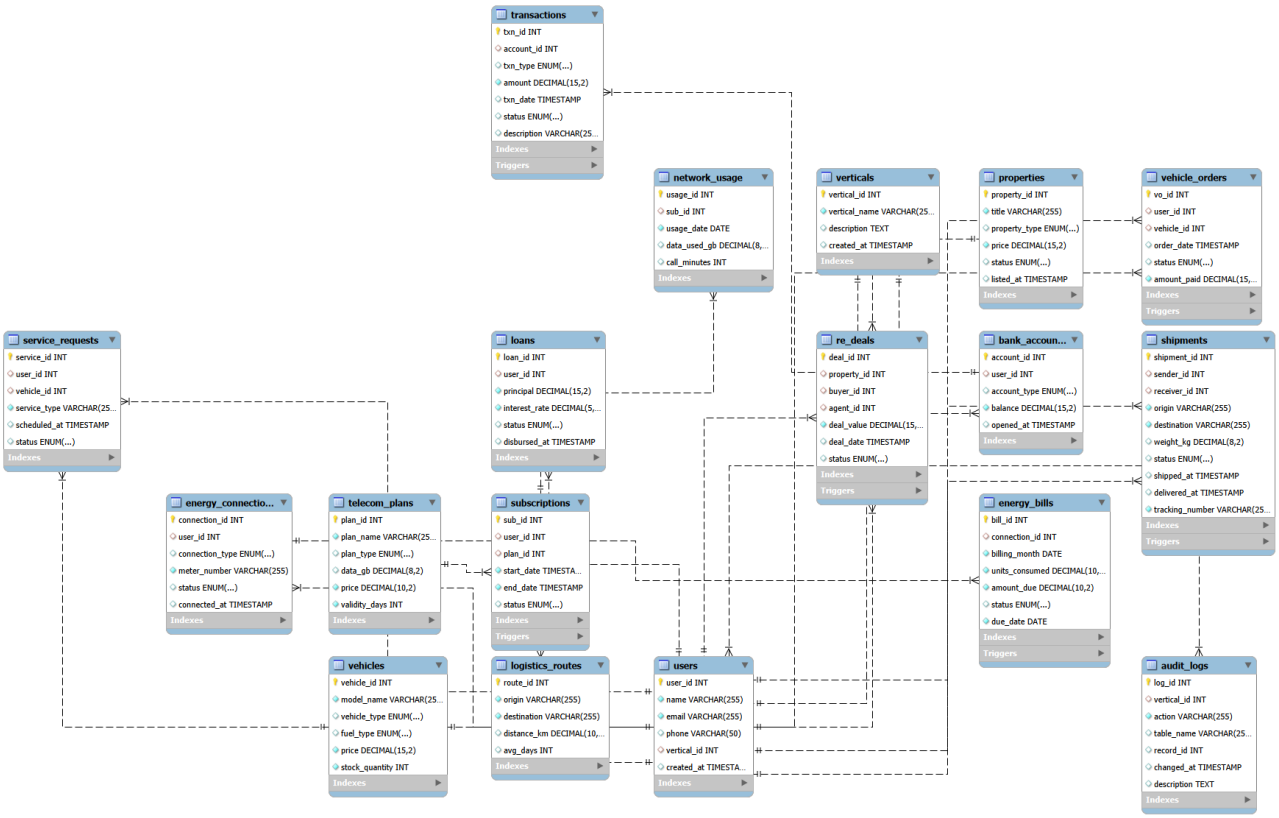

Figure 2: Relational Schema Diagram — MySQL Workbench

IMPLEMENTATION

8.1 Description of Tools and Technologies Used

Tool / Technology	Purpose
MySQL 8.0	Primary relational database for all tables and queries
MySQL Workbench	Schema design, ER diagram creation, and query testing
Node.js	JavaScript runtime environment for backend server
Express.js	Web framework for handling APIs and routing
MySQL2	Database connectivity and query execution
React.js	Frontend library for building user interface
Tailwind CSS	Styling and responsive design
Recharts	Data visualization using charts
Axios	API communication between frontend and backend
jsPDF, html2canvas	Client-side PDF generation for reports

8.2 Authentication Flow

1. **Login:** Admin enters username and password. The system verifies credentials (hardcoded check).
2. **Access Control:** On successful login, access to all dashboards and modules is granted.
3. **Session Handling:** Login status is maintained on the client side for accessing protected routes.
4. **Logout:** User logs out, and session is cleared from the application.

REFERENCES

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts*. McGraw-Hill.
2. MySQL 8.0 Reference Manual: `https://dev.mysql.com/doc/refman/8.0/en/`
3. Node.js Documentation: `https://nodejs.org/en/docs/`
4. Express.js Documentation: `https://expressjs.com/`
5. React.js Documentation: `https://react.dev/`
6. Tailwind CSS Documentation: `https://tailwindcss.com/docs`
7. Recharts Documentation: `https://recharts.org/en-US/`
8. Axios Documentation: `https://axios-http.com/docs/intro`
9. mysql2 npm package: `https://www.npmjs.com/package/mysql2`
10. jsPDF Documentation: `https://github.com/parallax/jsPDF`
11. html2canvas Documentation: `https://html2canvas.hertzen.com/`
12. MySQL Workbench Guide: `https://dev.mysql.com/doc/workbench/en/`